

Computer Networks I

Data Link Control Protocols (Medium Access Control Protocols) (Channel Partitioning, Taking Turns)

Amitangshu Pal

Computer Science and Engineering
IIT Kanpur

Multiple access links, protocols

Two types of “links”:

- Point-to-point
 - Point-to-point link between Ethernet switch, host
- Broadcast (shared wire or medium)
 - Old-school Ethernet
 - 802.11 wireless LAN, 4G/5G, satellite



humans at a cocktail
party
(shared air, acoustical)



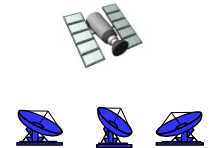
shared wire (e.g.,
cabled Ethernet)



shared radio: 4G/5G



shared radio: WiFi



shared radio: satellite

Multiple access protocols

- Single shared broadcast channel
- Two or more simultaneous transmissions by nodes:
 - **Collision** if node receives two or more signals at the same time

Multiple access protocol

- Distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
 - Communication about channel sharing must use channel itself!
 - ~~no out of band channel for coordination~~
-

An ideal multiple access protocol

Given: multiple access channel (MAC) of rate R bps

Desired rate:

1. When one node wants to transmit, it can send at rate R
 2. When M nodes want to transmit, each can send at average rate R/M
 3. Fully decentralized:
 - No special node to coordinate transmissions
 - No synchronization of clocks, slots
 4. Simple
-

MAC protocols: taxonomy

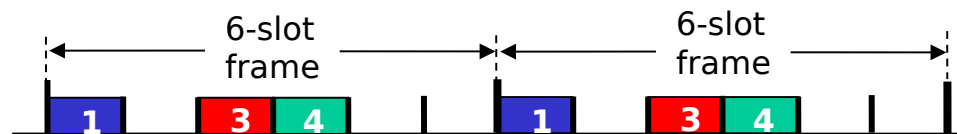
Three broad classes:

- **Channel partitioning**
 - Divide channel into smaller “pieces” (time slots, frequency, code)
 - Allocate piece to node for exclusive use
 - **Taking turns”**
 - Nodes take turns, but nodes with more to send can take longer turns
 - **Random access**
 - Use randomization for handling collisions
 - “Recover” from collisions
-

Channel partitioning protocols: TDMA

TDMA: Time division multiple access

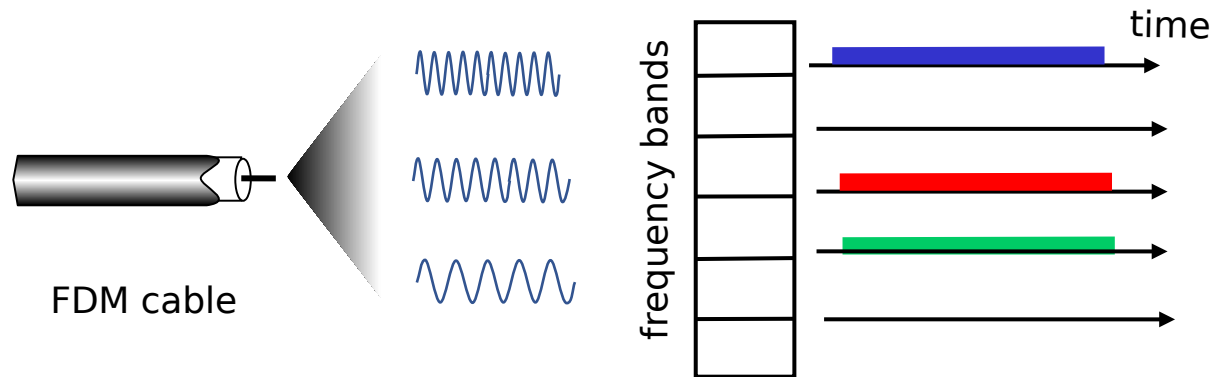
- Access to channel in “rounds”
- Each station gets fixed length slot (length = packet transmission time) in each round
- Unused slots go idle
- Example: 6-station scenario, 1,3,4 have packets to send, slots 2,5,6 idle



Channel partitioning protocols: FDMA

FDMA: Frequency division multiple access

- Channel spectrum divided into frequency bands
- Each station assigned fixed frequency band
- Unused transmission time in frequency bands go idle
- Example: 6-station scenario, 1,3,4 have packet to send, frequency bands 2,5,6 idle



MAC protocols: taxonomy

Three broad classes:

- **Channel partitioning**

- Divide channel into smaller “pieces” (time slots, frequency, code)
- Allocate piece to node for exclusive use

- **Taking turns”**

- Nodes take turns, but nodes with more to send can take longer turns

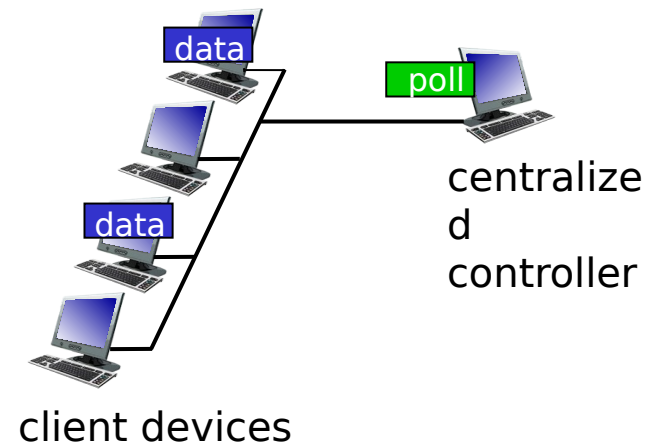
- **Random access**

- Use randomization for handling collisions
 - “Recover” from collisions
-

“Taking turns” MAC protocols

Polling:

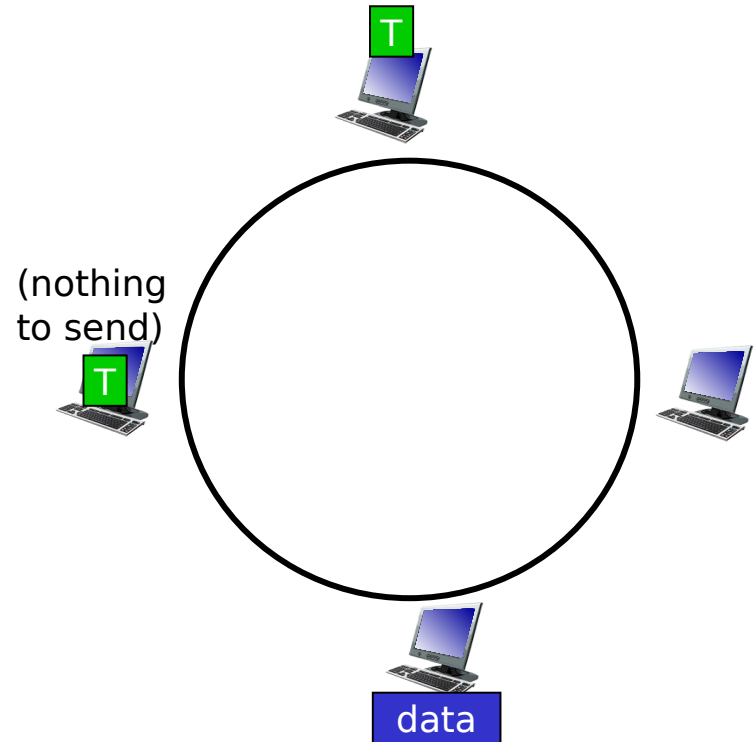
- Centralized controller “invites” other nodes to transmit in turn
- Typically used with “dumb” devices
- Concerns:
 - Polling overhead
 - Latency
 - Single point of failure (master)
- Bluetooth uses polling



“Taking turns” MAC protocols

Token passing:

- Control **token** message explicitly passed from one node to next, sequentially
 - Transmit while holding token
- Concerns:
 - Token overhead
 - Latency
 - Single point of failure (token)



Summary

□ Multiple Access Control:

- Channel partitioning, by time, frequency or code
 - TDMA, FDMA
 - Taking turns
 - Polling from central site, token passing
-